

Maximum Likelihood

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In this document, the maximum likelihood estimation (MLE) for the Ising and Potts models used in the `spininfer` library is defined and derived. First, the gradient derivations are presented for the Ising model and subsequently generalized to the Potts model.

1 Ising Model

This section provides a derivation of the gradient used in maximum likelihood estimation for the Ising model. The relevant equations are:

$$H(\boldsymbol{\sigma}) = -\sum_i h_i \sigma_i - \sum_i \sum_{j < i} J_{ij} \sigma_i \sigma_j, \quad (1)$$

$$p(\boldsymbol{\sigma}) = \frac{1}{Z} e^{-H(\boldsymbol{\sigma})} = \frac{1}{Z} \exp \left(\sum_i h_i \sigma_i + \sum_i \sum_{j < i} J_{ij} \sigma_i \sigma_j \right), \quad (2)$$

$$Z = \sum_{\boldsymbol{\sigma}} e^{-H(\boldsymbol{\sigma})}, \quad (3)$$

where Z is the partition function and $\boldsymbol{\sigma} \in \{-1, 1\}^N$ represents a system of N spins.

The average log-likelihood function for a dataset $\mathcal{D} = \{\boldsymbol{\sigma}^{(1)}, \dots, \boldsymbol{\sigma}^{(M)}\}$ consisting of M independent configurations is:

$$\mathcal{L}(\mathbf{h}, \mathbf{J}) = \frac{1}{M} \sum_{k=1}^M \ln p(\boldsymbol{\sigma}^{(k)} \mid \mathbf{h}, \mathbf{J}). \quad (4)$$

Expanding the log-likelihood yields:

$$\frac{\partial \mathcal{L}(\mathbf{h}, \mathbf{J})}{\partial h_i} = \frac{1}{M} \sum_{k=1}^M \left(-\frac{\partial H(\boldsymbol{\sigma}^{(k)})}{\partial h_i} \right) - \frac{\partial \ln Z}{\partial h_i}. \quad (5)$$

Using the identities:

$$\frac{\partial \ln Z}{\partial h_i} = \frac{1}{Z} \sum_{\boldsymbol{\sigma}} \sigma_i e^{-H(\boldsymbol{\sigma})} = \langle \sigma_i \rangle, \quad (6)$$

$$\frac{\partial H(\boldsymbol{\sigma})}{\partial h_i} = -\sigma_i, \quad (7)$$

and substituting (6) and (7) into (5), the gradient with respect to the local field is:

$$\frac{\partial \mathcal{L}(\mathbf{h}, \mathbf{J})}{\partial h_i} = \frac{1}{M} \sum_{k=1}^M \sigma_i^{(k)} - \langle \sigma_i \rangle = \langle \sigma_i \rangle_{\mathcal{D}} - \langle \sigma_i \rangle. \quad (8)$$

Similarly, differentiating the log-likelihood with respect to the coupling J_{ij} ($j < i$):

$$\frac{\partial \mathcal{L}(\mathbf{h}, \mathbf{J})}{\partial J_{ij}} = \frac{1}{M} \sum_{k=1}^M \left(-\frac{\partial H(\boldsymbol{\sigma}^{(k)})}{\partial J_{ij}} \right) - \frac{\partial \ln Z}{\partial J_{ij}}. \quad (9)$$

Using the corresponding identities:

$$\frac{\partial \ln Z}{\partial J_{ij}} = \frac{1}{Z} \sum_{\boldsymbol{\sigma}} \sigma_i \sigma_j e^{-H(\boldsymbol{\sigma})} = \langle \sigma_i \sigma_j \rangle, \quad (10)$$

$$\frac{\partial H(\boldsymbol{\sigma})}{\partial J_{ij}} = -\sigma_i \sigma_j, \quad (11)$$

we obtain:

$$\frac{\partial \mathcal{L}(\mathbf{h}, \mathbf{J})}{\partial J_{ij}} = \frac{1}{M} \sum_{k=1}^M \sigma_i^{(k)} \sigma_j^{(k)} - \langle \sigma_i \sigma_j \rangle = \langle \sigma_i \sigma_j \rangle_{\mathcal{D}} - \langle \sigma_i \sigma_j \rangle. \quad (12)$$

Equations (8) and (12) define the gradients used in the update rules for the moment matching objective.

2 Potts Model

For a system of N variables where each spin takes values in a discrete alphabet $\sigma_i \in \{1, \dots, q\}$, the Hamiltonian of the Potts model is:

$$H(\boldsymbol{\sigma}) = - \sum_i h_i(\sigma_i) - \sum_{i < j} J_{ij}(\sigma_i, \sigma_j), \quad (13)$$

where $h_i(a)$ is the local field for state a at site i , and $J_{ij}(a, b)$ is the pairwise coupling between site i in state a and site j in state b .

The probability of a configuration $\boldsymbol{\sigma}$ is given by the Boltzmann distribution:

$$p(\boldsymbol{\sigma}) = \frac{1}{Z} \exp \left(\sum_i h_i(\sigma_i) + \sum_{i < j} J_{ij}(\sigma_i, \sigma_j) \right), \quad (14)$$

with the partition function Z :

$$Z = \sum_{\boldsymbol{\sigma}} e^{-H(\boldsymbol{\sigma})}. \quad (15)$$

The average log-likelihood function for a dataset $\mathcal{D} = \{\boldsymbol{\sigma}^{(1)}, \dots, \boldsymbol{\sigma}^{(M)}\}$ is:

$$\mathcal{L}(\mathbf{h}, \mathbf{J}) = \frac{1}{M} \sum_{k=1}^M \ln p(\boldsymbol{\sigma}^{(k)} \mid \mathbf{h}, \mathbf{J}). \quad (16)$$

which we aim to maximize. Now, the gradient of the log-likelihood for a site with a given state $h_i(a)$ is:

$$\frac{\partial \mathcal{L}(\mathbf{h}, \mathbf{J})}{\partial h_i(a)} = \frac{1}{M} \frac{\partial \sum_{k=1}^M \ln p(\boldsymbol{\sigma}^{(k)} \mid \mathbf{h}, \mathbf{J})}{\partial h_i(a)}, \quad (17)$$

which can be decomposed into:

$$\frac{\partial \mathcal{L}(\mathbf{h}, \mathbf{J})}{\partial h_i(a)} = \frac{1}{M} \sum_{k=1}^M \left(-\frac{\partial H(\boldsymbol{\sigma}^{(k)})}{\partial h_i(a)} \right) - \frac{\partial \ln Z}{\partial h_i(a)}. \quad (18)$$

Once again the following identities can be derived:

$$\frac{\partial \ln Z}{\partial h_i(a)} = \langle \delta_{\sigma_i, a} \rangle, \quad (19)$$

$$\frac{1}{M} \sum_{k=1}^M \left(-\frac{\partial H(\boldsymbol{\sigma}^{(k)})}{\partial h_i(a)} \right) = \frac{1}{M} \sum_{k=1}^M \delta_{\sigma_i^{(k)}, a} = \langle \delta_{\sigma_i, a} \rangle_{\mathcal{D}}, \quad (20)$$

which yield the full derivative:

$$\frac{\partial \mathcal{L}(\mathbf{h}, \mathbf{J})}{\partial h_i(a)} = \langle \delta_{\sigma_i, a} \rangle_{\mathcal{D}} - \langle \delta_{\sigma_i, a} \rangle. \quad (21)$$

A similar approach is followed for the couplings, where the resulting derivative is:

$$\frac{\partial \mathcal{L}(\mathbf{h}, \mathbf{J})}{\partial J_{ij}(a, b)} = \frac{1}{M} \sum_{k=1}^M \left(-\frac{\partial H(\boldsymbol{\sigma}^{(k)})}{\partial J_{ij}(a, b)} \right) - \frac{\partial \ln Z}{\partial J_{ij}(a, b)}. \quad (22)$$

Now, these two parts can be rewritten as:

$$\frac{1}{M} \sum_{k=1}^M \left(-\frac{\partial H(\boldsymbol{\sigma}^{(k)})}{\partial J_{ij}(a, b)} \right) = \frac{1}{M} \sum_{k=1}^M \delta_{\sigma_i^{(k)}, a} \delta_{\sigma_j^{(k)}, b} = \langle \delta_{\sigma_i, a} \delta_{\sigma_j, b} \rangle_{\mathcal{D}}, \quad (23)$$

$$\frac{\partial \ln Z}{\partial J_{ij}(a, b)} = \langle \delta_{\sigma_i, a} \delta_{\sigma_j, b} \rangle, \quad (24)$$

yielding the gradient

$$\frac{\partial \mathcal{L}(\mathbf{h}, \mathbf{J})}{\partial J_{ij}(a, b)} = \langle \delta_{\sigma_i, a} \delta_{\sigma_j, b} \rangle_{\mathcal{D}} - \langle \delta_{\sigma_i, a} \delta_{\sigma_j, b} \rangle. \quad (25)$$

Together (21) and (25) give the update rules for the Moment Matching objective for the Potts model.